

TULGA, Turkey

WMO: 171970

Lat: 38.350N

Lon: 38.250E

Elev: 931

StdP: 90.63

Time Zone: 3.00 (E03)

Period: 06-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-7.0	-5.1	-14.0	1.3	-2.1	-12.0	1.5	-1.0	7.9	2.8	6.4	2.4	2.1	180	0.431

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	13.9	37.6	17.6	36.2	17.6	34.9	17.5	20.5	32.5	19.5	32.2	18.8	31.8	3.0	50

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
16.2	12.9	25.9	14.9	11.9	25.0	13.9	11.1	24.1	63.3	32.7	59.8	32.3	57.3	31.9	25.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.3	5.9	4.8	-9.5	39.8	2.5	1.2	-11.3	40.6	-12.8	41.3	-14.2	42.0	-16.0	42.9		
			-10.3	21.7	2.3	1.9	-12.0	23.1	-13.3	24.2	-14.6	25.3	-16.2	26.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	14.9	1.5	4.3	8.7	13.6	18.6	24.3	27.7	28.2	22.9	16.1	8.5	3.3	
	DBStd	9.74	3.53	3.71	3.56	3.30	3.22	2.71	2.32	2.29	3.32	3.12	3.13	3.43	
	HDD10.0	772	264	162	67	8	0	0	0	0	0	1	63	208	
	HDD18.3	2243	522	394	299	143	37	0	0	0	6	82	293	466	
	CDD10.0	2551	0	2	27	117	268	428	549	564	387	189	19	1	
	CDD18.3	979	0	0	0	3	46	178	291	306	143	13	0	0	
	CDH23.3	11737	0	0	0	31	444	1955	3724	3972	1495	115	0	0	
	CDH26.7	5938	0	0	0	1	117	873	2065	2233	636	12	0	0	
Wind	WSAvg	1.9	1.5	1.6	2.2	2.1	2.2	2.3	2.3	2.1	2.0	1.6	1.3	1.3	
Precipitation	PrecAvg	362	43	45	41	47	43	13	2	2	10	36	41	40	
	PrecMax	498	91	87	108	97	158	58	12	10	38	86	94	100	
	PrecMin	245	6	10	2	11	6	0	0	0	0	7	0	0	
	PrecStd	58	21	20	27	22	29	13	3	3	10	24	25	25	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.2	16.9	20.8	26.1	31.8	36.4	39.2	39.2	35.8	28.1	19.8	14.1	
		MCWB	6.4	9.3	11.5	13.1	16.2	17.9	17.5	18.0	15.9	15.2	12.2	8.2	
	2%	DB	9.0	13.8	18.8	24.0	29.7	34.5	37.8	37.9	34.0	26.1	17.5	10.8	
		MCWB	5.2	8.1	10.3	12.5	15.4	17.7	17.3	18.0	16.3	14.6	10.1	6.7	
	5%	DB	7.5	11.8	17.0	22.2	27.9	33.0	36.2	36.6	32.1	24.5	15.9	9.2	
		MCWB	4.3	6.9	9.4	12.0	15.2	17.1	17.4	17.8	16.4	14.4	9.4	6.0	
	10%	DB	6.2	9.9	15.1	20.4	25.9	31.3	34.8	35.2	30.4	22.7	14.0	7.9	
		MCWB	3.5	5.7	8.6	11.5	14.7	16.7	17.4	17.7	16.0	13.5	9.0	5.6	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.2	9.9	13.0	16.0	18.2	20.6	21.6	22.8	19.8	17.4	13.6	9.4
			MCDB	9.3	15.8	18.3	22.4	27.9	31.7	32.5	31.9	30.1	23.9	17.9	12.9
		2%	WB	5.9	8.6	11.2	14.2	17.0	19.2	20.2	20.7	18.7	16.0	11.6	8.0
			MCDB	8.3	12.8	16.8	20.8	26.8	31.4	32.3	33.2	29.7	23.4	14.5	9.6
5%		WB	4.8	7.4	10.1	13.1	16.0	18.5	19.3	19.7	17.7	15.1	10.4	6.8	
		MCDB	6.9	11.0	15.7	19.7	25.0	30.3	32.0	33.0	29.0	22.3	14.0	8.6	
10%		WB	3.9	6.2	8.9	12.2	15.2	17.7	18.6	18.9	16.7	14.3	9.4	5.8	
		MCDB	5.7	9.2	14.5	18.6	23.8	29.1	31.7	32.9	28.2	20.7	13.3	7.3	
Mean Daily Temperature Range	5% DB	MDBR	6.3	7.9	9.5	10.9	12.1	13.3	13.7	13.9	13.5	10.9	9.5	6.7	
		MCDBR	8.0	10.8	12.0	13.5	14.1	14.3	15.0	14.9	15.3	13.5	12.1	8.6	
		MCWBR	5.2	6.4	6.3	6.2	5.7	5.2	5.2	5.3	5.9	6.3	6.7	5.6	
	5% WB	MCDBR	6.7	9.5	10.8	11.3	12.8	13.4	13.2	13.7	13.6	11.4	9.7	6.3	
		MCWBR	4.8	6.1	6.2	6.0	5.7	5.7	5.7	6.2	6.5	6.0	6.1	4.5	
Clear-Sky Solar Irradiance	taub	0.331	0.365	0.407	0.457	0.427	0.372	0.362	0.378	0.348	0.390	0.345	0.327		
	taud	2.244	2.154	2.083	1.984	2.134	2.326	2.360	2.295	2.368	2.197	2.294	2.302		
	Ebn at Noon	828	844	844	823	857	904	910	887	898	806	807	806		
	Edh at Noon	100	125	149	175	154	127	122	127	111	119	95	89		
All-Sky Solar Radiation	RadAvg	2.13	3.09	4.25	5.32	6.53	7.89	8.03	7.11	5.82	3.91	2.66	1.87		
	RadStd	0.29	0.51	0.40	0.50	0.40	0.36	0.22	0.23	0.24	0.30	0.31	0.39		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (9 neighbors)	N/A	N/A	N/A	N/A	-0.88	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page